



DEEP MUSCLE ACTIVITY MEASUREMENTS USING NEAR-INFRARED SPECTROSCOPY

Bachelor Thesis

Submitted in partial fulfillment of the requirements for the Bachelor's
Degree Programme in Engineering – Electrical Engineering and Information
Technology with emphasis on automation and medical technology

Authors

Ali Haidar Al-Bedani,
Yoan Tutunarov,
Henrik Bernhard Elvegård,
Asem Ilyas

Supervisor

Peyman Mirtaheri

A thesis presented for the degree of
Bachelor of Electronics Engineering

Oslo Metropolitan University, 2024
Faculty of Technology, Art and Design
Department of Mechanical, Electronic
and Chemical Engineering

Acknowledgments

We would first like to thank Oslo Metropolitan University for providing us with the unique opportunity to engage in a real-world experiment, supporting us financially and academically throughout the development of our deep muscle monitoring device using near-infrared spectroscopy. A special thank you to our supervisor, Prof. Peyman Mirtaheri, for his unwavering support, invaluable expertise, and for always being available to answer our questions. We also extend our heartfelt thanks to Håvard Engell and GaitLine for the insightful contributions and understanding of the psychological aspects of our research. Their input has supported the growth of our project.

Abstract

Biomedical engineering merges engineering methods with medical science to tackle healthcare challenges, leading to breakthroughs like advanced prosthetics and physiological monitoring technologies. Despite its rapid advancement and significant contributions, the field remains relatively small, with limited public awareness, possibly due to the complexity of integrating engineering with medical sciences. Our project builds on electronic engineering and in-depth anatomical and physiological knowledge, venturing into a relatively unexplored area. By leveraging existing resources and advancing innovative solutions, we aim to develop a novel solution for measuring the blood supply to the sural nerve, addressing a gap in current biomedical engineering applications. This attempt not only showcases the field's potential for significant innovations but also emphasizes the importance of interdisciplinary research in unlocking new medical engineering frontiers. Hopefully, this project will fill this gap and give a clearer understanding of the sural nerve's importance. We believe that this can further be used to contribute to people's well-being.

Group Members' Contributions

Group Members	Written Contribution	Technical Contribution
Ali Haidar Al-Bedani	Pages 10, 18, 24-26, 27-30 & 43	Circuit and PCB design, make PCB and solder components, developing and editing Project report
Henrik Elvegård	Pages 12, 13, 20, 21, 33 - 35, 38	Circuit design and coding
Asem Ilyas	Pages 1, 11, 12, 14-18, 23-24. 38-42	Design and assembling final product package
Yoan Tutunarov	Pages 2, 8, 9, 11, 14, 16-20, 30-32 & 43-44	Circuit design and coding

Contents

Acknowledgments	1
Abstract	2
Group Members' Contributions	3
Summary	7
1 Introduction	8
1.1 Objective	8
1.2 Goals	8
1.3 Project structure	8
2 Background and purpose	9
2.1 Methods for measurements	9
2.1.1 LDF:	9
2.1.2 NIRS:	10
2.1.3 Ultrasonography:	12
2.2 Background and summary of state of status	13
2.3 Choice of project goals and boundaries with justification	14
2.4 Method specifications	15
3 Theory	16
3.1 The anatomical structure of the sural nerve	16
3.2 Blood	17
3.3 Blood supply to the sural nerve	18
3.4 Cause of sural nerve's activation	18
3.5 Light and tissue interaction	18
3.5.1 Absorption	19
3.5.2 Reflection	20
3.5.3 Modified Beer-Lambert law	20
3.6 Specifications for measurements	21
3.6.1 Wavelengths	21
3.6.2 Depth	22
4 Methods, Material, Tools for implementation	23
4.1 Materials	23
4.2 Method/Equipment	23
4.3 Implementation Details	27
4.3.1 LED Drivers	27
4.3.2 Transimpedance Amplifier	28
4.3.3 Analog solution	30
4.3.4 Digital solution	32
4.4 Block diagram of the system	36
4.5 Circuit sketches	37

4.6	Test procedure for system tests / experiments / user tests	38
5	Results	39
5.1	Description of the "product" with references to theory	39
5.2	Experimental data	40
5.3	Discussion	42
6	Conclusions and Further Work	43
	References	45

List of Figures

1	Gantt chart	9
2	Electromagnetic Specter	12
3	Illustration of our LED placement and the skin	15
4	Illustration of sural nerve	17
5	(Sorbellini, Rucco, Rinaldi, 2018)	19
6	(Alwazzan, 2020)	22
7	PCB circuit schematic	25
8	(Amplifier)	30
9	(2. order sallen key high-pass)	31
10	(2.order sallen key low-pass)	31
11	(Arduino, n.d.)	32
12	Arduino code for IIR lowpass filter	34
13	simulink	35
14	Arduino Lock-in amplifier del1	35
15	Arduino Lock-in amplifier del2	36
16	Arduino Lock-in amplifier del3	36
17	Arduino Lock-in amplifier del4	36
18	Block diagram of device with both analog and digital approach .	37
19	Circuit sketch of the transmitting circuit	37
20	Circuit sketch of the reciving circuit	38
21	Construction approach	38
22	Product case	39
23	Open case	40
24	Casting of silicone case for probe	41
25	Using 5Hz simulated signal	42
26	Same circuit filtering simulated signal in figure 25	42
27	Test conducted for functionality of LED drivers	43

Summary

This bachelor thesis explores the integration of electronic engineering with anatomical and physiological knowledge to address a specific gap in biomedical engineering: deep muscle activity measurements using near-infrared spectroscopy (NIRS). Recognizing the sural nerve's critical role in sensing mechanical stimuli and maintaining balance, the project aims to develop a prototype device using NIRS. This device will measure changes in the blood supply to the sural nerve, providing real-time, non-invasive monitoring of oxygenated and deoxygenated hemoglobin levels.

The research is structured into three main phases: investigation, documentation, and execution. Initially, an extensive literature review and methodological analysis were conducted, identifying NIRS as the most suitable approach due to its cost-effectiveness and capability to provide precise measurements. The project emphasizes simplicity by utilizing dual-wavelength NIRS, balancing feasibility with effective monitoring.

Throughout the development phase, continuous testing and rigorous troubleshooting ensured the reliability of each circuit component. The project's scope is intentionally limited to constructing a baseline device and offering preliminary insights, setting a foundation for future, more detailed research.

By clarifying the interaction between mechanosensors within the sural nerve and the resultant hemodynamic changes, this thesis contributes to a deeper understanding of peripheral nerve health and lays the groundwork for advancements in medical diagnostics and wearable health technology. This interdisciplinary approach underscores the potential for significant innovations in biomedical engineering, enhancing both academic knowledge and practical healthcare applications.

1 Introduction

1.1 Objective

Incorrect foot placement during walking can lead to injuries and discomfort in the lower body and back over time. This issue impacts many adults, adversely affecting their quality of life and posing significant societal costs. Mastering a proper walking technique and maintaining a healthy foot posture are critical. It has been determined to delve into the involved nervous system and its inherent physiology to comprehend how even minor alterations in foot positioning can influence the foot and the entire body, specifically the sural nerve, which serves as the cornerstone of our project.

The sural nerve, an integral component of the peripheral nervous system, plays a pivotal role in sensing touch, temperature, vibration, and pain in the foot and lower leg. This sensory function is crucial for maintaining balance and facilitating precise foot placement during movement (Grose, 2022). Recent research emphasizes the significance of mechanosensors in the skin, which translate mechanical stimuli into neural signals. Mechanosensitive ion channels are at the forefront of this conversion process, underscoring their importance in the tactile sensation (Handler & Ginty, 2021). The sural nerve's accessibility and its role in detecting mechanical stimuli position it as a key area of study for understanding peripheral nerve health and functionality (Grose, 2022; Miniato & Nedeff, 2023).

1.2 Goals

The objective of this project is to construct a prototype device that measures changes in the blood supply to the sural nerve when it is activated, subsequently producing a signal output. Furthermore, a comprehensive and organized report will be prepared to support our development of the actual device.

1.3 Project structure

Our project is primarily structured into three phases: investigation, documentation, and execution. Due to the distinct and intricate nature of our task, investigation plays a pivotal role in our project. Extensive research was conducted both before and during the project to comprehend the problem we're addressing.

The documentation serves as the blueprint for the final product, as it supplies facts, logical reasoning, and technical specifications. Without clearly defined project stages and precise specifications, achieving a satisfying result becomes almost unattainable.

The approach we've adopted for this project begins with research, proceeds with building the documentation, and swiftly moves into the execution of the device. It's crucial to continue documenting the project concurrently with the development of the circuit and device, rather than postponing it until the end.



Figure 1: Gantt chart

2 Background and purpose

2.1 Methods for measurements

Monitoring the activities of the sural nerve holds considerable interest for enhancing our understanding of the intricacies involved in the functioning of mechanosensors, especially those uniquely situated in the heel skin. This focus is particularly pertinent as it provides valuable insights into the complex interactions and responses of these sensory mechanisms. Such understanding is critical, not only for advancing our knowledge of neurological functions but also for implications in designing better therapeutic and diagnostic strategies that target the sensory pathways. This exploration could potentially lead to significant improvements in treating and managing conditions that affect the sensory nerves.

Hence, we'll be exploring three significant techniques for measuring alterations in blood supply: near-infrared spectroscopy, Ultrasound, and Doppler Flowmetry. To make an informed decision, it's essential to understand the theoretical principles behind these methods.

2.1.1 LDF:

Laser Doppler Flowmetry (LDF) is a real-time technique for assessing blood flow in microvascular systems. It has been widely applied in various fields, including dermatology, dentistry, and clinical diagnostics. LDF is based on the Doppler effect, where a coherent laser light beam interacts with moving red blood cells, causing a shift in frequency that can be measured to determine blood

flow characteristics. This technique offers a range of technical specifications crucial for its effectiveness and application, including responsiveness, volume, depth, sensitivity, and selectivity.

Responsiveness: LDF is known for its high responsiveness, providing real-time monitoring of microcirculation blood flow. It is capable of detecting rapid changes in blood flow almost instantaneously, making it suitable for applications where continuous monitoring is essential (Fredriksson et al., 2012).

Volume: The measurement volume in LDF refers to the size of the tissue area where blood flow is measured. Typically, LDF measures blood flow in a small, localized volume, around 1 to 3 cubic millimeters (mm^3). However, the exact volume can vary based on specific device configurations and the tissue's optical properties (Fredriksson et al., 2012; Metzger et al., 2011; Fredriksson et al., 2009).

Depth: The depth of penetration for LDF typically ranges from 1.5 to 2 millimeters for standard 2 milliWatt He-Ne laser light, making it effective for superficial tissues. Newer devices may vary the wavelength between 600 and 700 nm depending on the tissue's absorption properties to optimize depth penetration. Depth discrimination is particularly important in applications like skin microcirculation monitoring, where distinguishing between superficial and deeper blood flow is crucial (Metzger et al., 2011).

Sensitivity and Selectivity: LDF is highly sensitive, capable of detecting minimal blood flow, including low perfusion levels in microvascular beds. Its sensitivity is further enhanced by using multi-wavelength lasers and self-mixing detection methods, which allow for more accurate detection of the Doppler shift caused by moving red blood cells. While selectivity in LDF typically refers to its ability to measure microvascular blood flow specifically, advancements aim to enhance its selectivity by providing more targeted depth measurements and possibly distinguishing between different layers of tissue microcirculation (Metzger et al., 2011).

2.1.2 NIRS:

NIRS represents a cost-effective, non-invasive optical methodology extensively utilized across various medical disciplines. It operates on the principle of infrared (IR) light penetration through superficial anatomical layers, including the skin, subcutaneous fat, and cranial bone. NIRS specifically harnesses wavelengths in the near-infrared spectrum, typically from 700 nm to 900 nm, which are optimally suited for deep tissue penetration and exhibit differential absorption characteristics when interacting with oxyhemoglobin and deoxyhemoglobin. This selective absorption is pivotal as it facilitates the quantification of hemoglobin oxygenation and total hemoglobin volume changes. The interaction of IR light with these hemoglobin forms results in either absorption or scattering, which NIRS devices measure to obtain critical data on tissue oxygen saturation, hemoglobin concentration variations, and oxygen consumption rates in cerebral and muscular tissues. Consequently, NIRS is frequently employed in both clinical evaluations and biomedical research, proving to be an invaluable

instrument for real-time and dynamic assessment of physiological functions.

Responsiveness: NIRS demonstrates exceptional responsiveness to alterations in blood oxygenation and hemoglobin concentrations within biological tissues. The absorption intensity of light, as detected by NIRS sensors, exhibits a direct correlation with the levels of oxyhemoglobin and deoxyhemoglobin present. Consequently, even subtle shifts in hemoglobin content or oxygen saturation, triggered by physiological dynamics such as changes in oxygen demand or vascular flow, are accurately captured by this technology. This high degree of responsiveness renders NIRS particularly adept at real-time monitoring of neural and muscular tissues. By facilitating the observation of real-time physiological changes linked to metabolic and vascular activities, NIRS allows clinicians and researchers to track fluctuations inherently associated with vital biological processes, including those of the respiratory and cardiac cycles. This capability is crucial for the nuanced study and management of conditions that affect the cerebral and muscular systems.

Volume: The optical density recorded by NIRS exhibits a proportional relationship to the volumetric presence of blood within the examined tissue. Enhanced blood flow phases, such as systole, are characterized by an augmented presence of hemoglobin-rich blood, thereby elevating the absorption of near-infrared light and consequentially increasing the optical density measurement. In contrast, during diastole—a phase marked by reduced blood flow—the optical density values correspondingly diminish. This proportional relationship between optical density and blood volume enables NIRS to serve as a highly effective instrument for the quantitative assessment of fluctuations in blood flow and hemoglobin concentration, which are indicative of various physiological conditions. Thus, NIRS provides critical insights into the dynamic vascular changes associated with different phases of cardiac activity.

Depth: The depth of NIRS depend on various factors. The most important factor is the wavelength of the diode we choose. The scientific light specter in Figure 2 shows different wavelengths, for infrared wavelengt between 700 nm to 900 nm is applicable. Power of light source is another factor that affect the depth NIRS can penetrate.

High-power NIRS systems are often used for measuring deoxygenation in deep muscle tissues during exercise and have a depth of up to 3cm in tissue (Koga et al., 2015).

Sensitivity and selectivity: NIRS offers high sensitivity and selectivity when proper techniques and models are applied. Sensitivity is optimized through calibration and wavelength selection, while selectivity is achieved by distinguishing unique spectral features of analytes using multivariate analysis and feature selection methods.

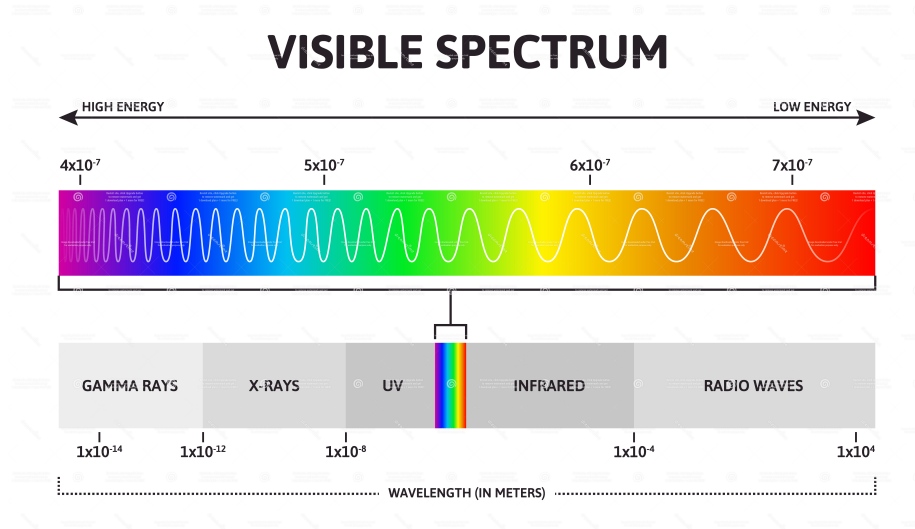


Figure 2: Electromagnetic Specter

2.1.3 Ultrasonography:

Ultrasound refers to acoustic waves with a frequency higher than human hearing (20kHz). External ultrasonography is an ultrasound-based diagnostic imaging technique used to visualize subcutaneous tissues. Ultrasound machines utilize a transducer composed of piezoelectric crystals. When an electric signal is applied, these crystals vibrate at a high frequency, generating ultrasound waves in the 2 - 18 MHz (Carovac et al., 2011). Some of the transmitted waves are reflected to the transducer as they encounter tissues of different densities. The intensity of the returning echo determines the brightness of the image on the screen. Various tissues are displayed in shades of gray, ranging from white to black, based on their respective densities (Shah et al., 2011). One of the main applications of ultrasonographical imaging is examining blood (Kremkau, 2021). This is achieved using a technique called Doppler sonography. As the name implies, this technique is based on the physical principles of the doppler effect for soundwaves. The technique allows for measuring and visualizing the direction and speed of blood flow (Carovac et al., 2011). There are various types of doppler methods, such as spectral doppler, power doppler and color doppler. The velocity of blood flow relative to the probe is displayed in a color spectrum using the color-Doppler method. Data processing enables ultrasound machines to combine the grayscale image and color information. The blood flowing toward the probe is colored red and the blood flowing away from the probe is colored in blue. When the speed of a bloodstream exceeds the Nyquist frequency a phenomenon called aliasing occurs. Aliasing causes an inversion

of highspeed blood jets. Using continuous doppler highspeed blood flow can be measured, but the exact location of the highest velocity will be unknown (Kremkau, 2020).

Responsiveness: Ultrasound machines yield a scan or frame in a fraction of a second and repeat this process several times in a second to produce real-time imaging (Kremkau, 2020). With the use of continuous doppler, one part of the transducer is always transmitting the ultrasound while the other part is always receiving the reflected waves. With pulsed doppler the transducer transmits a pulse of waves and sends out a new pulse when the reflected pulse is registered. This limits the time resolution for pulsed doppler but makes it possible to locate the exact sample value (Dalen Brabrand, 2016).

Volume: In ultrasonography, the sample volume is the place where the blood flow is measured (Dalen Brabrand, 2016).. It is also known as sample length and is typically between 0.33 and 0.5 mm. The sample volume can be adjusted (Oglat et al., 2018). Confusing presentations can appear when more than one motion of blood flow is included in the sample volume. This can be solved by narrowing the sample volume (Kremkau, 2020).

Depth: Ultrasound has a limited penetration depth depending on the frequency. Low-frequency ultrasound can penetrate deeper into tissues but with lower resolution. High-frequency ultrasound provides higher resolution but doesn't penetrate as deeply. For measuring blood flow in deeper vessels, low-frequency Doppler examinations may be more suitable (Dalen Brabrand, 2016).

Sensitivity and selectivity: Ultrasonography is sensitive to blood flow due to its ability to detect and provide information about the presence or absence of flow, the velocity of flow and the character of flow. The resolution of sonographic imaging is dependent on the framerate and resolution which both is reduced with increased angle and depth. Power Doppler is independent of angle and aliasing but can not provide information about direction, speed and character of flow. This method is sensitive to low-speed blood flow, making it suitable for measuring small blood vessels (Dalen Brabrand, 2016; Kremkau, 2020).

2.2 Background and summary of state of status

Since we will measure the environment around the sural nerve, the most suitable place is the lateral side above the ankle. It is also close to the heel, where sural nerve stimulation occurs and thus measures the blood supply. The other key factor in this area is that not much can cause noise in the measured signal. Cost, complexity and expected size of the device are factors considered. NIRS is the most cost efficient method that we have discussed as the most appropriate for this project. LDF is a solution that could be suited for our output signal, but the pros of its responsiveness and depth have their downside of its excessive amount of unwanted signal and noise. The last method that we had available was ultrasonography, a method that is more flexible and can

Table 1: Methods

Method	Time	Volume	Depth	Sensitivity	Selectivity
LDF	Real-time	1-3 mm ³	1.5-2 mm	High for microvascular flow	Good for microvascular flow
NIRS	Real-time	Sample volume: Measurement depth-dependent	wavelength dependent/Power dependent	Good for microvascular flow	Good for microvascular flow
US	Real-time	0.33 - 0.5 mm ³	Frequency dependent	Good for microvascular flow	Good for microvascular flow

easily be modified. The reason this is not a method we used further is because the cost to implement this method is exceedingly high, and the change in depth often affects the image resolution. The flexibility we get is good for one part of the project, but a change in this method affects other aspects, making it more inaccurate.

2.3 Choice of project goals and boundaries with justification

The primary objective of this project is to develop a Near-Infrared Spectroscopy (NIRS) device utilizing dual light sources to distinguish between oxygenated and deoxygenated hemoglobin in the tissue surrounding the sural nerve as shown in Figure 3. This methodology leverages the differential absorption properties of hemoglobin at specific wavelengths, thereby enabling non-invasive, real-time monitoring of the nerve’s microvascular blood supply. The choice of using only two wavelengths—rather than a more complex multi-wavelength system—stems from considerations of simplicity, feasibility within the project’s limited timeframe, and extensive research of dual-wavelength for discerning key forms of hemoglobin. The scope of this bachelor thesis is intentionally limited to constructing a baseline device and providing preliminary insights into hemodynamic conditions around the sural nerve. This approach not only mitigates the practical constraints associated with a more elaborate setup but also establishes a foundational framework for future, more detailed research. By delineating these boundaries, the project maintains manageable objectives that adhere to the academic requirements and available resources, while setting a ro-

bust groundwork for subsequent investigations that could expand on this initial exploration.

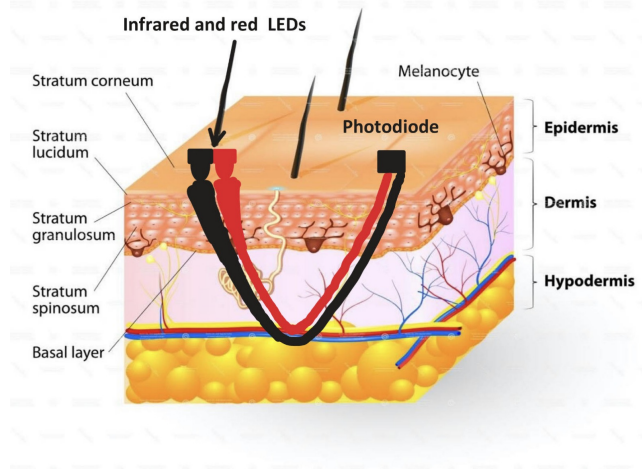


Figure 3: Illustration of our LED placement and the skin

2.4 Method specifications

The initiation of our research entailed a thorough examination of the scientific literature, undertaken to discern the most efficacious methodology for our device. This comprehensive review was strategically aligned with the overarching goals of our project. Subsequent to this rigorous academic investigation, Near-Infrared Spectroscopy (NIRS) emerged as the superior approach for attaining the outcomes we envisaged. The preference for NIRS was significantly predicated on its cost-effectiveness relative to alternative methodologies available in the field.

During the developmental phase of the project, a meticulous process of continuous testing was implemented for each circuit component. This was essential to ensure that any discrepancies or errors were promptly identified and rectified, thereby mitigating any potential impact on the integrity of the overall output signal. The methodological implementation followed a systematic procedure, which involved the initial design of the component, followed by rigorous testing of the output signal. Should the output not align with the predetermined expectations, the process necessitated a troubleshooting cycle until the signal quality met our stringent criteria. This iterative process of testing and correction was imperative, particularly considering the constrained time frame of the project, which introduced significant challenges in thoroughly ensuring the quality and reliability of the signal.

The principal objective of this project is to establish a robust baseline that encapsulates both the data acquired and the functionality of the device. This

foundational work is intended to lay the groundwork for subsequent research and development, providing a reliable platform for further detailed investigations. In achieving this, the project aims to contribute a seminal framework that supports the advancement of knowledge and technology in the field.

3 Theory

3.1 The anatomical structure of the sural nerve

The sural nerve functions exclusively as a sensory nerve responsible for transmitting sensory signals from the skin. It originates from two smaller nerves within the sural nerve complex: the medial cutaneous nerve, which branches off from the tibial nerve, and the lateral cutaneous sural nerve, which branches off from the common peroneal nerve. The union forming the sural nerve is located in the distal third of the posterior leg. Running alongside the small saphenous vein, the sural nerve courses between the two heads of the gastrocnemius muscle as it descends towards the dorsolateral aspect of the ankle. From there, it travels posteriorly between the calcaneus and the lateral malleolus before extending lateral outwards towards the fifth toe (Miniato Nedeff, 2023).

The sural nerve is located under our skin in the subcutaneous tissue. Our skin is built up by stratum cornea (dead skin), epidermis, dermis, and subcutaneous tissue, also called hypodermis. According to a study on Korean adults, the thickness of the dermis and epidermis together is between 0.521 mm and 1.977 mm, with a mean of 1.249 mm (Lee Hwang, 2002). Another study on pregnant women measured the thickness of the epidermis, dermis, and subcutaneous tissue to be 4.6 ± 0.4 mm (Yanagisawa et al., 2019). At last, an article for anatomical landmarks of the ankle block states that the sural nerve in the ankle is between 4–6mm perpendicularly deep to the skin (Nimana et al., 2023). This information equips us with a reliable estimate of the sural nerve’s depth under the skin surface, in addition to the skin’s thickness. Pinpointing the exact position and depth of the sural nerve would be highly intricate. Consequently, this project aims to perform measurements in the area surrounding the blood supply of the sural nerve.



Figure 4: Illustration of sural nerve

3.2 Blood

This project focuses on the changes in deoxygenated hemoglobin (Hbo2) and oxygenated hemoglobin (Hb). When a nerve is activated, there is an increase in metabolic concentration. Metabolic activity is necessary for maintaining the electric potential across cell membranes, neurotransmitter release, and signal propagation along the nerve fibers. The increased metabolic activity in the nerve and surrounding tissue leads to a higher demand for oxygen to support cellular respiration and ATP production, which are essential for sustaining nerve function and signal. For the oxygen demand, the active surrounding tissue extracts oxygen from the bloodstream as blood flows through the capillaries in the region. Oxygen diffuses from capillaries into the the interstitial fluid and then into the cells, where it is utilized for energy production.

The oxygen is extracted from the bloodstream by metabolically active tissue, and the concentration of Hb in the blood decreases. Hb releases oxygen to the tissues, transitioning into deoxyhemoglobin in the process.

Hbo2 is the form of hemoglobin that has released its oxygen molecules and is ready to transport carbon dioxide back to the lungs for exhalation. Consequently, the concentration of Hbo2 increases in the blood, leaving the activated nerve region. This reflects the proportion of hemoglobin that has delivered its oxygen to the tissue.

3.3 Blood supply to the sural nerve

The movement of ions during nerve activation, due to mechanical stimuli, requires energy. This energy is supplied in the form of ATP (Adenosine Triphosphate). ATP is often referred to as the 'energy currency' of the cell because it powers many of the cellular processes, including the movement of ions across the cell membrane.

The production of ATP, however, requires oxygen, which is where the role of blood oxygenation comes into play. Oxygen is carried to the cells by the blood, where it is used in a process called cellular respiration to produce ATP. During this process, oxygen (O_2) is consumed and carbon dioxide (CO_2) is produced as a waste product. The CO_2 is then removed from the cells and transported back to the lungs by the blood, leading to deoxygenation.

Consequently, the activation of the sural nerve and the subsequent ion concentration changes lead to an increased demand for oxygen, thereby affecting blood oxygenation and deoxygenation. This is why monitoring the levels of blood oxygenation and deoxygenation can provide us with valuable information about the functioning of the sural nerve and the metabolic activities that are associated with its activation.

3.4 Cause of sural nerve's activation

As we walk, our heel ideally strikes the ground in a vertical motion. The first point of contact on the heel coincides with the location of the sural nerve. At this point, mechanosensory cells within our skin react to the mechanical stimuli. A mechanosensitive ion-channel known as PIEZO2 is identified as the primary mechanosensor in these cells. It initiates the bioelectrical process, in which mechanical input is converted into a cellular response. The application of mechanical force on the cell membrane causes cations (positively charged ions) to flow through the ion channel. This leads to a depolarization of the cell membrane in neurons and triggers an action potential. (Salzer Boehm, 2019; Szczot et al., 2021). When an action potential (nerve impulse) is generated, it stimulates the next neuron. This allows the signal to reach the spinal cord and brain via the sural nerve and the rest of the nervous system (Chawla Sahni, 2016).

We understand nerve impulses influence blood flow to the nerves and throughout the body. Consequently, when the sural nerve is stimulated, it's reasonable to anticipate alterations in its blood supply. This is why we're interested in gauging the variations in the sural nerve's blood supply when it's activated compared to when it's not.

3.5 Light and tissue interaction

Figure 5 shows a detailed view of light penetration through various skin layers. As seen on the left side, skin comprises four primary layers: stratum corneum, epidermis, dermis, and subcutaneous tissue. The dermis is further di-

vided into sublayers, namely, papillary dermis, upper blood net dermis, reticular dermis, and deep blood net dermis, each containing varying degrees of vascular concentration. The color lines in the picture illustrate the penetration depth of different light wavelengths through the skin. Blue and green light penetrate the epidermal and papillary dermal skin layers. Yellow to red light penetrates more deeply, reaching the dermal vascular structure. Near-infrared (NIR) light extends to the parts of the subcutaneous tissue.

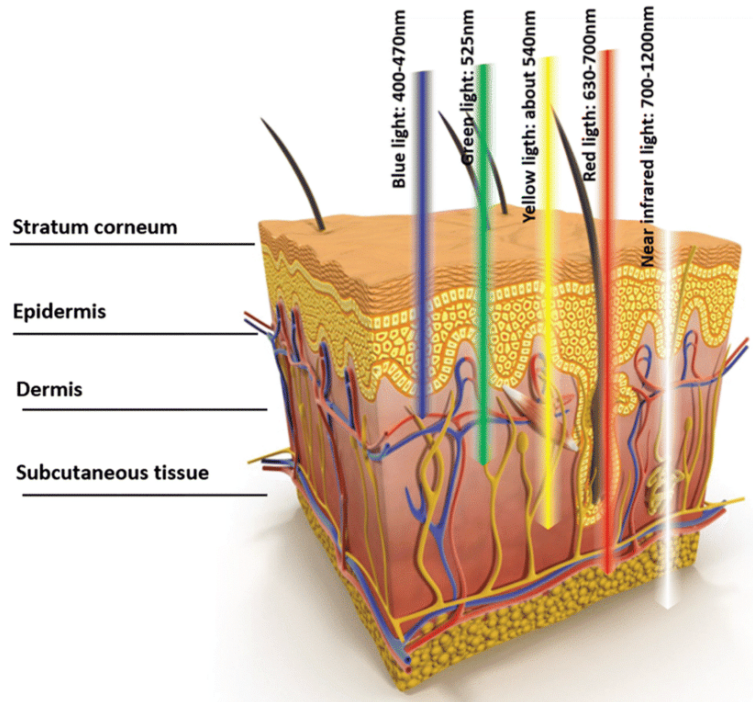


Figure 5: (Sorbellini, Rucco, Rinaldi, 2018)

3.5.1 Absorption

Skin and tissue absorbance and scattering are key biomedical optics and imaging factors. These properties influence how light interacts with biological tissues, such as skin, and are critical for optical measurement techniques. Absorbance refers to how much light is absorbed by the tissue. Different tissues and substances within tissues (like water, melanin, and hemoglobin) absorb light differently. For instance, melanin in the epidermis absorbs more light at shorter wavelengths, while hemoglobin in blood absorbs light at specific wavelengths, depending on whether it's oxygenated or deoxygenated. Scattering refers to how light is dispersed or scattered as it travels through tissue. Scattering occurs when light encounters interfaces between different tissue types or structures within the

tissue, like cells and fibers. Scattering often increases at shorter wavelengths, known as Rayleigh scattering (Kyriacou Chatterjee, 2022).

3.5.2 Reflection

In near-infrared spectroscopy (NIRS), reflection refers to the phenomenon where light, after being emitted from a source, interacts with a material (biological tissue) and then bounces back to be detected by a sensor. This is often called diffuse reflection or backscattering.

When near-infrared light enters a tissue, it doesn't just pass straight through. It gets absorbed and scattered, or bounced around, within the tissue. Some of this scattered light returns to the surface and contributes to the detected signal.

The detected reflected light contains valuable information about the tissue it interacts with. For instance, by analyzing the pattern and intensity of this reflected light, we can make inferences about the oxygenation level of the tissue, which is crucial in medical diagnostics and research (Ferrari Quaresima, 2012; Scholkmann et al., 2014).

3.5.3 Modified Beer-Lambert law

The color of a medium occurs due to chromophores, which are molecules that absorb and transmit light at certain wavelengths. The absorption of light by chromophores reduces emerging light intensity. A fundamental principle in spectroscopy called the Beer-Lambert law describes how the change in light intensity is related to the concentration of chromophores, the thickness of the solution, and the extinction coefficient.

$$A = \log \left[\frac{I_0}{I} \right] = \varepsilon * c * d \quad (1)$$

A is the medium's absorption, I_0 and I are the intensity of the source light and detected light, respectively, ε is the molar extinction coefficient that describes the chromophore's optical characteristics, c is the concentration of the chromophore, and d is the distance between the source and detector (optode spacing).

Biological tissue is a highly scattering medium; therefore, the reduction in emerging light intensity occurs not only due to absorption but is also significantly impacted by scattering. Multiple scattering events can be experienced by photons, contingent upon the optical properties of tissue layers and the wavelength of the light source. This results in an effective optical path length surpassing the optode spacing. Hence, a differential path length factor (DPF) is incorporated into the beer-lambert law to correlate the actual optical path length with the source-to-detector separation. This modified version of beer-lambert law is the foundation for NIRS, which calculates a chromophore's concentrations (Owen-Reece et al., 1999).

$$A = \varepsilon * c * d * DPF \quad (2)$$

Changes in the concentration of a chromophore, such as oxygenated and deoxygenated hemoglobin, can be calculated when measurements of changes in light intensity are made. The concentration is calculated by rearranging the modified Beer-Lamberts law.

$$\Delta C = \frac{\Delta A}{\varepsilon d * DPF} \quad (3)$$

To measure the changes in concentrations of oxygenated and deoxygenated hemoglobin, two or more light sources with distinct wavelengths are required. The wavelengths used are 760nm and 850nm. The wavelengths lie within the optical window, a part of the near-infrared spectrum where other significant chromophores have a low absorption factor. Taking into account that two light sources are used, the beer-lamberts law can be expressed as:

$$\Delta A_{\lambda 1} = (\varepsilon_{HHb\lambda 1} \Delta HHB + \varepsilon_{HbO_2\lambda 1} \Delta HbO_2) d * DPF \quad (4)$$

$$\Delta A_{\lambda 2} = (\varepsilon_{HHb\lambda 2} \Delta HHB + \varepsilon_{HbO_2\lambda 2} \Delta HbO_2) d * DPF \quad (5)$$

Solving for the concentration of oxygenated and deoxygenated hemoglobin yields:

$$\Delta HbO_2 = \frac{\Delta A_{\lambda 1} \varepsilon_{HHb\lambda 2} - \Delta A_{\lambda 2} \varepsilon_{HHb\lambda 1}}{(\varepsilon_{HbO_2\lambda 1} \varepsilon_{HHb\lambda 2} - \varepsilon_{HbO_2\lambda 2} \varepsilon_{HHb\lambda 1}) d * DPF} \quad (6)$$

$$\Delta HHB = \frac{\Delta A_{\lambda 2} \varepsilon_{HbO_2\lambda 1} - \Delta A_{\lambda 1} \varepsilon_{HbO_2\lambda 2}}{(\varepsilon_{HbO_2\lambda 1} \varepsilon_{HHb\lambda 2} - \varepsilon_{HbO_2\lambda 2} \varepsilon_{HHb\lambda 1}) d * DPF} \quad (7)$$

(Abay Kyriacou, 2015)

Selecting a dpf that is best suited to the measurement parameters is crucial for achieving accurate results. A study focused on estimating dpf for skin showed that the optode separation and wavelength of the light source greatly impact the dpf. As the optode separation increases the dpf also increases and as the wavelength increases the dpf decreases. Furthermore, the study showed that the dpf is negligibly affected by the amount of melanin in the skin (Althobaiti, 2023).

3.6 Specifications for measurements

3.6.1 Wavelengths

To keep track of the oxygen levels in the tissues surrounding the sural nerve, we need to look at both the increase and decrease of oxygen values. To do this, information from measurements taken at two separate wavelengths is needed because of the way hemoglobin, a protein in red blood cells, absorbs light.

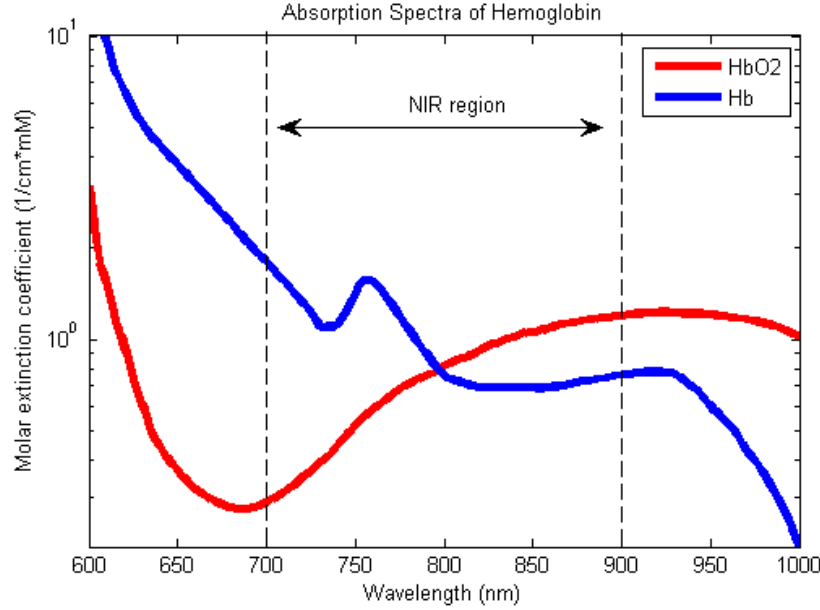


Figure 6: (Alwazzan, 2020)

Hemoglobin's role is to transport oxygen from the lungs to different parts of the body and then bring carbon dioxide back to the lungs. Figure 6 illustrates the absorption of light by hemoglobin at various wavelengths. The red line represents oxygenated hemoglobin, while the blue line represents deoxygenated hemoglobin. Looking at the graph, it's clear that between 600nm and around 800nm, hemoglobin without oxygen absorbs more light than hemoglobin with oxygen. The point where the red and blue lines intersect is known as the "isosbestic point", where both types of hemoglobin absorb light equally. After this point, the roles reverse at higher wavelengths.

The selection of light sources for this project was greatly guided by this graph. The decision was to use two infrared LEDs at wavelengths of 740 nm and 870 nm. We selected these particular wavelengths because we believe they provide sufficient information. They are neither too close nor too far from the "isosbestic point", making them ideal for our purpose.

3.6.2 Depth

Depth is crucial in addition to wavelength. We've gathered data from various studies about the position of the sural nerve beneath the skin's surface. Based on this information, we estimate that the nerve is located 3 to 6 mm beneath the skin in the ankle area. We've chosen to focus on a depth of 4 mm because this would cover the region affected by stimulation of the sural nerve.

Our detector will be placed at the same level as our light source, so it relies on the light's reflection. It's important to note that different types of light and wavelengths penetrate to different depths, and this is also influenced by the material they interact with.

Importantly, the distance between the light source and the detector is critical for determining how deep the light can reach. According to Artinis Medical Systems, a Dutch company, the depth reached by NIR spectroscopy is roughly half the distance between the light source and the detector. Therefore, this distance will be set at 8mm for our purposes (Artinis Medical Systems, 2022).

4 Methods, Material, Tools for implementation

4.1 Materials

The device incorporates a range of low-cost components as detailed in Table 2, each carefully selected to meet specific measurement specifications. It is powered by two commonly available 9V batteries, chosen for their reliability and accessibility. Light detection is optimized through a photodiode that matches the wavelengths of the dual infrared LEDs used as the light source. These LEDs emit at 720 nm and 870 nm, wavelengths specifically chosen to correspond with the absorption spectrum of hemoglobin.

The design features an assortment of resistors ranging from 92 ohms to 1 M Ω , selected to fine-tune the circuit's time constant, thus enhancing the device's measurement response and stability. Capacitors are similarly chosen based on their charging times to maintain circuit efficiency. At the heart of the device's digital operations is the Arduino Mega 2560, selected for its robust processing capabilities and significant memory capacity, crucial for managing the LEDs and depicted in the block diagram Figure 18. This microcontroller is integral to the device's signal processing and filtering functionalities, increasing its versatility and application scope.

Connectivity for analog components is facilitated through a standard breadboard, chosen for its adaptability and ease of modifications—an essential feature for ongoing developmental adjustments. Digital elements are systematically arranged on a custom-designed double-sided breadboard that optimizes spatial efficiency and simplifies connections, reflecting the successful completion and thoughtful engineering of the device.

4.2 Method/Equipment

The methodology employed in the development of this device is rooted in fundamental engineering principles, characterized by systematic testing and iterative adjustments based on the diagnostic outcomes derived from these tests. These procedural details are illustrated in figure 21.

Table 2: Materials

Component	Value	Quantity
Power supply	5V	1
Photodiode	-	1
LED	640nm - 870nm	2
Resistor	82 - 1M	24
Capasitor	10pF - 1uF	5
Arduino Mega2560	-	1
Breadbord	-	1
Double side custom PCB	-	2
Case	-	1

For a methodical and comprehensible implementation, the device’s circuitry was meticulously divided into discrete segments. This segmentation was strategic, designed to streamline the assembly process and enhance troubleshooting efficiency. Each individual segment was subjected to exhaustive testing, utilizing a signal generator to simulate various input signals. An oscilloscope was employed as the pivotal diagnostic tool throughout these tests. This instrument played an essential role in verifying that the signals produced were being accurately processed according to the predefined specifications.

The device’s design incorporated two distinct development pathways: an analog approach and a digital approach. In adopting the digital methodology, a Printed Circuit Board (PCB) of the circuit was created. This phase was critical in laying the foundational electrical framework for the device.

PCBs are fundamental components in electronic devices, providing mechanical support and electrical connections for electronic components. This section details the hands-on PCB creation process, highlighting the traditional photolithographic technique used. The first step in creating a PCB is designing the circuit schematic. For this, we used EasyEDA to map out all the electrical connections and components. The schematic serves as a blueprint, ensuring all components are correctly placed and interconnected, as shown in figure 7. The

with a thin layer of copper, is prepared. The printed transparency sheet is placed over the copper-clad board and then exposed to UV light. The UV light passes through the transparent areas of the sheet, hardening the photoresist on the copper surface below. It is critical to conduct this step in complete darkness, except for the controlled UV exposure, to prevent premature exposure of the photoresist. After UV exposure, the board undergoes a development process. This involves washing the board in a solution of water mixed with a developer, sodium carbonate. This solution removes the unexposed photoresist, revealing the copper beneath, while the exposed photoresist remains, protecting the underlying copper.

The next step is etching, where the board is submerged in an etching solution, ferric chloride. This solution dissolves the unprotected copper, leaving behind only the copper traces covered by the hardened photoresist. The etching process typically takes about 10 minutes but can vary depending on the concentration of the etchant and the thickness of the copper layer. After etching, the board is thoroughly washed in water to remove any residual etching solution. Post-etching, the PCB is trimmed to remove any excess material, ensuring the board is the correct size and shape for its intended application. Holes are then drilled into the PCB for through-hole components. This step requires precision drilling to ensure the components fit correctly and make proper electrical connections.

The PCB is then cleaned to remove any remaining photoresist and other residues. This step is crucial for preparing the board for soldering. The last step before soldering on components is applying flux to the PCB to clean the copper surfaces and prevent oxidation, which facilitates the soldering process by ensuring good electrical connections. The traditional photolithographic method of PCB creation is a thorough process that requires attention to detail and precision. Understanding this manual process provides valuable insights into the fundamentals of PCB fabrication, despite the availability of modern automated techniques.

Continuing with the digital approach, the Arduino IDE was employed to program an Arduino microcontroller. This programming was tailored to meet the specific requirements of our signal processing objectives, ensuring that the device performed according to the intended operational parameters.

To further refine the processing capabilities of the device and ensure its optimal functionality, MATLAB-Simulink was integrated into the development process. The choice of MATLAB-Simulink was influenced by its extensive array of analytical tools and advanced processing options, which are indispensable for complex data analysis and system modeling. To effectively integrate Simulink with our existing Arduino-based system, a specialized add-on was utilized. This add-on was crucial for translating the detailed Simulink circuit designs into a format that was compatible with the Arduino, thus ensuring a seamless transition and functional integrity of the system.

Table 3: Tools and equipment

Tool /equipment
Oscilloscope
Function generator
Arduino IDE
Matlab - Simulink
Easy eda

4.3 Implementation Details

The system built in this project is made of a transmitting circuit and a receiving circuit. the transmitting circuit is responsible for controlling the emitting light. It consists of two LEDs, a driver circuit and an Arduino microcontroller. The solution to the receiving circuit consists of two approaches: an analog approach and a digital approach. Both approaches use a photodiode and a transimpedance amplifier. The analog approach uses further analog amplification and filtering in multiple steps to gain the dc component of the signal. The digital approach utilizes a digital lock-in amplifier algorithm programmed on the Arduino. Combined with a digital lowpass filter, the lock-in amplifier provides the dc component of the detected signal.

4.3.1 LED Drivers

LED drivers are essential components in the operation of LED lighting systems. They perform several critical functions that ensure the reliable and efficient operation of LEDs. LEDs are current-driven devices, meaning their brightness is directly related to the current passing through them. LED drivers provide a stable current source, preventing fluctuations that could otherwise lead to variations in brightness or color. Without proper current regulation, LEDs can be damaged by excessive current, leading to reduced lifespan or immediate failure. LED drivers incorporate current limiting features that protect LEDs from over-current conditions, ensuring longevity and reliability. Additionally, LED drivers handle voltage regulation, accommodating variations in input voltage to ensure that LEDs receive the correct voltage.

To send a signal through an LED, a dedicated LED driver is required due to the forward voltage of the LED, denoted as V_f . This forward voltage reduces

part of the signal voltage V_s , where $V_{s, \text{out}} = V_{s, \text{in}} - V_f$. To maximize the effectiveness of the signal, a constant voltage supply that is equal to or greater than V_f is necessary. Therefore, an LED driver circuit is needed to combine the signal voltage, V_s , and the constant voltage, while limiting the current to the LED's optimal operational level.

The non-inverting voltage adder circuit utilizes an operational amplifier (op-amp) to sum the input voltages. The first input voltage, denoted as V_1 , is a signal that varies between 0V and 5V. The second input voltage, V_2 , is a constant voltage set at around 5V. The configuration of the op-amp in this circuit ensures that the output voltage, V_{out} , is the sum of the two input voltages, mathematically expressed as:

$$V_{\text{out}} = V_1 + V_2 \quad (8)$$

To protect the circuit from excessive current, a current-limiting resistor, R_L , is added at the output. The purpose of this resistor is to limit the current flowing through the load to a safe level. The value of the current-limiting resistor can be determined using Ohm's Law, which relates voltage, current, and resistance:

$$R_L = \frac{V_{\text{out}} - V_f}{I_{\text{max}}} \quad (9)$$

where I_{max} is the desired maximum current through the load and V_f is the forward voltage of the LED. By selecting an appropriate value for R_L , the output current can be kept within safe limits, thereby protecting the circuit from potential damage.

4.3.2 Transimpedance Amplifier

After sending the signals from the Arduino through the LED drivers to the LEDs, the photodiode will detect the signal. Since both the photodiode and LEDs operate using current, it is necessary to use a TIA to convert this current into a voltage. The reason for this conversion is that the analog-to-digital converter (ADC) on the Arduino can only read voltage, not current. The TIA is crucial in this setup because it performs several important functions:

1. **Current-to-Voltage Conversion:** The primary function of a TIA is to convert the current signal generated by the photodiode into a corresponding voltage signal. This conversion is essential because the photodiode generates a current proportional to the intensity of the light it receives, and this current must be translated into a voltage signal for further processing.
2. **Amplification:** The TIA amplifies the small current produced by the photodiode. Photodiodes typically generate very low current levels, often in the microampere range. The TIA amplifies this current to produce a measurable voltage that can be accurately read by the ADC.

3. **Impedance Matching:** The TIA provides an impedance matching function, ensuring that the high impedance of the photodiode does not affect the performance of the subsequent circuitry. By converting the high-impedance current signal into a low-impedance voltage signal, the TIA allows for more efficient signal processing and reduces the potential for noise and signal loss.
4. **Noise Reduction:** The TIA helps to minimize noise in the signal. By converting and amplifying the current signal close to the source (the photodiode), the TIA reduces the amount of noise that can be introduced into the signal path. This results in a cleaner and more accurate signal being sent to the ADC.

The use of a TIA in this application is essential for accurately converting and processing the signals received by the photodiode. This ensures that the Arduino's ADC can effectively read and interpret the signal, enabling precise control and monitoring of the LED system.

The implementation of the TIA involves connecting the photodiode to the inverting input of an operational amplifier (op-amp) with a feedback resistor, R_f , across the op-amp. The feedback resistor determines the gain of the TIA and, consequently, the amplitude of the output voltage. The output voltage, V_{out} , can be expressed as:

$$V_{out} = -I_{ph} * R_f \quad (10)$$

where I_{ph} is the current generated by the photodiode in response to the incident light. The negative sign indicates an inversion of the signal, which is typical in TIA configurations.

By carefully selecting the value of the feedback resistor, R_f , we can control the sensitivity and range of the TIA to match the expected current levels from the photodiode. This ensures that the output voltage falls within the input range of the Arduino's ADC, allowing for accurate digital representation and further processing. The implementation of the TIA involves connecting the photodiode to the inverting input of an operational amplifier (op-amp) with a feedback resistor, R_f , across the op-amp. The feedback resistor determines the gain of the TIA and, consequently, the amplitude of the output voltage. The output voltage, V_{out} , can be expressed as:

$$V_{out} = -I_{ph} * R_f \quad (11)$$

where I_{ph} is the current generated by the photodiode in response to the incident light. The negative sign indicates an inversion of the signal, which is typical in TIA configurations.

In addition to the feedback resistor, adding a capacitor in parallel with the resistor can introduce a low-pass filter into the TIA circuit. This configuration is useful for filtering out high-frequency noise and stabilizing the output signal. The combination of the feedback resistor R_f and the capacitor C_f forms an RC filter.

The cutoff frequency f_c of the low-pass filter is determined by the values of R_f and C_f , and can be calculated using the formula:

$$f_c = \frac{1}{2\pi R_f C_f} \quad (12)$$

By selecting appropriate values for R_f and C_f , we can set the cutoff frequency to filter out unwanted high-frequency components, ensuring that the TIA's output signal is clean and stable. This is particularly important in applications where the photodiode is exposed to ambient light fluctuations or electrical noise.

The addition of a capacitor in parallel with the feedback resistor in a TIA circuit enhances the circuit's performance by filtering out high-frequency noise, resulting in a more stable and accurate output signal. This improvement is crucial for ensuring reliable signal processing in photodiode-based systems.

4.3.3 Analog solution

Amplifiers

It's assumed that the optical signal reflected back to the photodiode is extremely weak, necessitating amplification to produce a signal that can be read. A single non-inverted amplifier was used in the photodiode circuit for this purpose. The circuit also included two resistors, with values of $68,000\Omega$ and $2,200\Omega$ respectively, resulting in an estimated gain of 31. It is important to note that amplification is a key element for improving the final output but was not always included in the test procedures. The amplifier's calculations were aided by Okawa Electric Design.

Op-amp circuit:

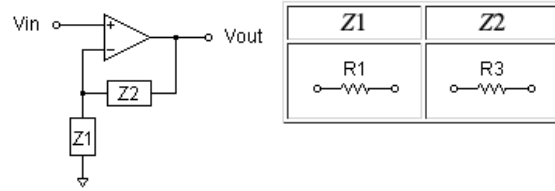


Figure 8: (Amplifier)

Filtering

Analog filters in signal processing are used to modify, or filter, the aspects of analog signals. They work on continuous signals and can enhance or attenuate a specific frequency range. Common types include low-pass, high-pass, band-pass, and band-stop filters. These filters are essential in applications like audio

and radio signal processing. The raw signal acquired from the NIRS method in this task is highly susceptible to disturbances. It can easily be affected by extraneous frequencies and noises, leading to a subpar output signal. Utilizing filters aids in eliminating unwanted frequencies and external interferences. Both filters use two resistors and two capacitors. The circuit incorporates two filters:

2. order Sallen Key High pass filter with a cut-off frequency at approximately 0.34Hz. The values of R1 and R2 are $1\text{M}\Omega$ and the values of C1 and C2 are $0.47\mu\text{F}$.

Sallen-Key High-pass Filter

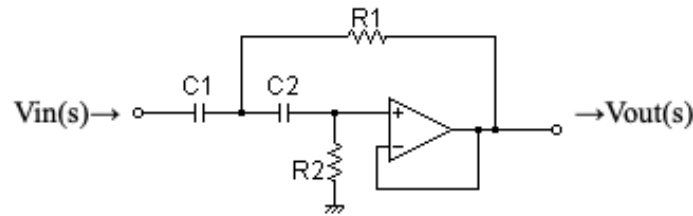


Figure 9: (2. order sallen key high-pass)

2. order Sallen Key Low pass filter with a cut-off frequency at approximately 1Hz. The values of R1 and R2 are $150\text{k}\Omega$ and the values of C1 and C2 are $1\mu\text{F}$.

Sallen-Key Low-pass Filter

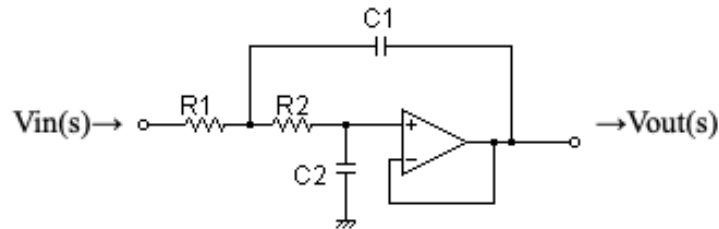


Figure 10: (2.order sallen key low-pass)

To compute the various components of the circuits, including filters and amplifiers, a digital electrical design tool was leveraged (Okawa Electric Design, n.d.).

4.3.4 Digital solution

In the process of creating a functional device to address our task, we transitioned from an analog to a digital solution. This shift was motivated by the enhanced possibilities, size-efficiency, and alignment with modern electronic solutions offered by a digital system.

Micro-controller

Serving as the system's central processing unit, the microcontroller governs the small LED circuit and processes the signal received from the photodiode. Our project employs the Arduino Mega 2560 (ATmega) microcontroller, which offers cost-effectiveness, efficiency, and compact size. It is managed using the Arduino IDE, which employs a language similar to C.

When compared to the Arduino Uno, a lower-tier microcontroller used in our initial stages, the ATmega presents enhanced power and larger capacity, making it ideal for more complex projects.

The Arduino Mega operates at 5V, meaning it can provide up to 5V to an external component. It boasts a total of 54 pins, 15 of which support Pulse Width Modulation (PWM), and has 16 analog inputs.

Furthermore, it comes with 256KB of Flash memory and operates at a clock speed of 16MHz (Arduino, n.d.).

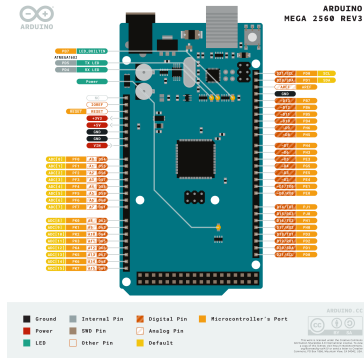


Figure 11: (Arduino, n.d.)

Lock-in amplifier

A lock-in amplifier is a circuit able to extract a small signal from significantly greater noise. It works by using phase-sensitive detection to gain the signal from the noise. The lock-in amplifier requires an external or internal reference signal with the same frequency and phase as the input signal. This signal is often a sine wave or a square wave. By using a multiplier or a phase-sensitive detector, the input signal is multiplied by the reference signal, exploiting the properties

of trigonometry and the fact that all waves can be represented as a sum of sines and cosines, the output is expressed as the product of the two signals. (source)

$$V_{out} = V_{in} V_{ref} \sin(\omega_{in} t + \theta) \sin(\omega_{ref} t) \quad (15)$$

$$V_{out} = \frac{V_{in} V_{ref}}{2} \cos((\omega_{in} - \omega_{ref}) t + \theta_{in} - \theta_{ref}) - \frac{V_{in} V_{ref}}{2} \cos((\omega_{in} + \omega_{ref}) t + \theta_{in} + \theta_{ref}) \quad (16)$$

By applying a lowpass filter at the output of the lock-in amplifier, the expression is simplified.

$$V_{out} = \frac{V_{in} V_{ref}}{2} \cos\theta \quad (17)$$

Digital filtering

Digital filtering is a key process in many systems and applications to manipulate or enhance signals. It involves the use of mathematical operations to modify an input signal in order to extract useful information or achieve a desired output.

Digital filters are typically designed to either enhance or attenuate certain frequencies present in the input signal. For example, they can be used to reduce noise, extract trends, or isolate specific frequency components.

There are two main types of digital filters:

1. Finite Impulse Response (FIR) filters: These filters use only input data to create an output and have a finite response to an impulse input. They are inherently stable and can have exactly linear phases.

2. Infinite Impulse Response (IIR) filters: These filters use both previous input and previous output data to compute the output. They can achieve a given filter characteristic using fewer coefficients than FIR filters, but they may have stability issues and their phase response is not always linear.

In this project, the digital filter was engineered using the Arduino IDE and Python, which were utilized to compute the filter coefficients. These coefficients play a crucial role in setting the cut-off frequency of the filter. Here, an IIR lowpass filter was employed, incorporating three distinct coefficients. It allows low-frequency components to pass while attenuating high-frequency components (Proakis Manolakis, 1996).

The ensuing code and coefficients were utilized:

```

1 // This file shows an example implementation of a low-pass filter on an Arduino.
2 // Note that there are many potential improvements to this code.
3
4 float xn1 = 0;
5 float yn1 = 0;
6 int k = 0;
7
8 void setup() {
9   Serial.begin(115200);
10 }
11
12 void loop() {
13   // Test signal
14   float t = micros()/1.0e5;
15   // Bytte ut med ingangssignal
16   //float xn = sin(2*PI*2*t) + 0.2*sin(2*PI*50*t);
17   float xn = analogRead(A0);
18   // Compute the filtered signal
19   float yn = 0.969*yn1 + 0.0155*xn + 0.0155*xn1;
20
21   delay(1);
22   xn1 = xn;
23   yn1 = yn;
24
25   if(k % 3 == 0){
26     // This extra conditional statement is here to reduce
27     // the number of times the data is sent through the serial port
28     // because sending data through the serial port
29     // messes with the sampling frequency
30
31     // Output
32     Serial.print(xn);
33     Serial.print(" ");
34     Serial.println(yn);
35   }
36   k = k+1;
37 }

```

Figure 12: Arduino code for IIR lowpass filter

Software tools

The initial implementation of the digital approach was conducted using the Arduino IDE, the primary software for programming Arduino microcontrollers. A program incorporating the lock-in algorithm and lowpass filtering was developed, as illustrated in Figures 14 - 17. In addition to using the Arduino IDE, a solution was also developed in MATLAB. MATLAB provides a visual modeling platform known as Simulink, which uses block diagrams to represent systems. Simulink offers numerous advantages over the Arduino IDE, particularly for more complex designs such as a lock-in amplifier. The choice of MATLAB-Simulink was influenced by its extensive array of analytical tools and advanced processing options, which are indispensable for complex data analysis and system modeling. The digital solution constructed in Simulink is depicted in Figure 13.

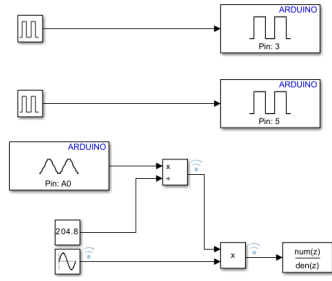


Figure 13: simulink

In the first part of the lock-in algorithm variables and constants are declared. Digital outputs for the LEDs and reference signals is set up in void setup. In void loop, digital outputs for the LEDs and reference signals is set up. The analog input pin connected to the photodiode is also set up and the serial monitor started with a baudrate of 115200.

```

1
2 const int led1 = 3;
3 const int led2 = 5;
4 const int ref1 = 2;
5 const int ref2 = 4;
6
7 const int pd = A0;
8 float measureVal;
9 float voltageVal;
10 float multival;
11
12 int state1;
13 int state2;
14
15 const int time = 500;
16
17 float xn1 = 0;
18 float yn1 = 0;
19 int k = 0;
20
21 void setup() {
22   Serial.begin(115200);
23   pinMode(ref1, OUTPUT);
24   pinMode(ref2, OUTPUT);
25   pinMode(led1, OUTPUT);
26   pinMode(led2, OUTPUT);
27   pinMode(pd, INPUT);
28 }

```

Figure 14: Arduino Lock-in amplifier del1

While the first LED is on and the corresponding reference signal is high, a measurement from the photodiode is made. The measured value is further converted to a range from 0 - 5 volts. The lowpass filter is then applied.

Through the if statement, the amount of data flowing through the serial port is limited as this greatly affect the sampling frequency. further down the code, the next part of the void loop starts. In this part, the first LED is set turned off while the other second is turned on. the first referencesignal is set low as the second reference signal is set high.


```

29
30 void loop() {
31
32     digitalWrite(ref1, 1);
33     digitalWrite(ref2, 0);
34
35     digitalWrite(led1, 1);
36     digitalWrite(led2, 0);
37
38     state1 = digitalRead(ref1);
39     state2 = digitalRead(ref2);
40
41     measureVal = analogRead(pd);
42     voltageVal = 5 * measureVal / 1023;
43     multiVal = voltageVal * state1;
44
45     float t = micros()/1.0e6;
46
47     float xn = multiVal;
48     float yn = 0.969*yn1 + 0.0155*xn + 0.0155*xn1;
49
50     delay(1);
51     xn1 = xn;
52     yn1 = yn;

```

Figure 15: Arduino Lock-in amplifier del2

```

53
54 if(k % 3 == 0){
55     Serial.print(xn);
56     Serial.print(" ");
57     Serial.println(yn);
58 }
59 k = k+1;
60
61 delay(time);
62
63 digitalWrite(ref1, 0);
64 digitalWrite(ref2, 1);
65
66 digitalWrite(led1, 0);
67 digitalWrite(led2, 1);
68
69 state1 = digitalRead(ref1);
70 state2 = digitalRead(ref2);
71
72 measureVal = analogRead(pd);
73 voltageVal = 5 * measureVal / 1023;
74 multiVal = voltageVal * state2;
75
76 float xn = multiVal;
77 float yn = 0.969*yn1z + 0.0155*xnz + 0.0155*xn1z;

```

Figure 16: Arduino Lock-in amplifier del3

```

78
79 delay(1);
80 xn1 = xn;
81 yn1 = yn;
82
83 if(k % 3 == 0){
84     Serial.print(xn);
85     Serial.print(" ");
86     Serial.println(yn);
87 }
88 k = k+1;
89
90 delay(time);
91
92 }
93

```

Figure 17: Arduino Lock-in amplifier del4

4.4 Block diagram of the system

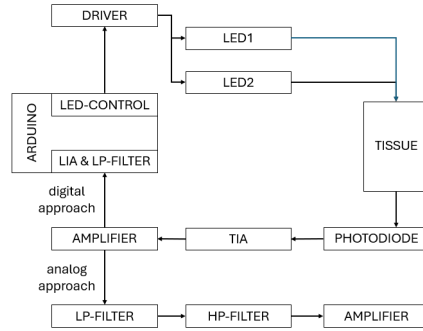


Figure 18: Block diagram of device with both analog and digital approach

4.5 Circuit sketches

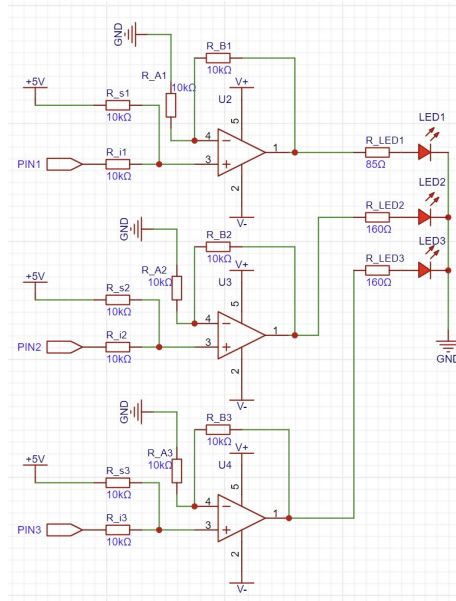


Figure 19: Circuit sketch of the transmitting circuit

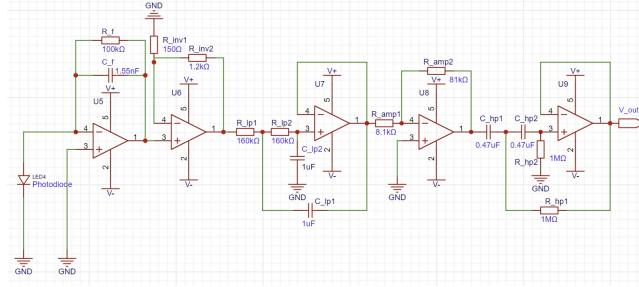


Figure 20: Circuit sketch of the receiving circuit

4.6 Test procedure for system tests / experiments / user tests

To ensure the system functioned as intended, a straightforward test procedure was implemented. This procedure is designed to verify that each part of the circuit operates correctly both individually and in conjunction with the preceding part of the circuit.

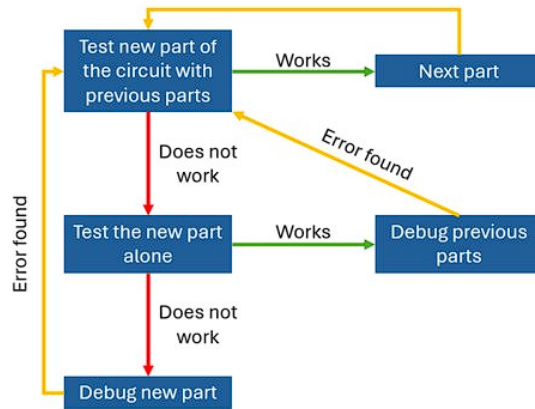


Figure 21: Construction approach

A signal generator was utilized during testing to produce a signal with highly specific frequency and amplitude. This capability is particularly useful for testing filters, as it allows verification that the filter's frequency response matches theoretical predictions. An oscilloscope was also essential during testing due to its ability to graph voltage over time. Furthermore, the oscilloscope was used to display the magnitude of frequencies by employing Fast Fourier Transform (FFT) analysis. The circuit was powered by a dual power supply during testing. By connecting the positive output of one power supply to the negative output

of the other, a stable ground was established.

5 Results

Due to unforeseen factors, comprehensive data collection using our device on the tissues in the body was not feasible. We encountered limitations that prevented the production of a functional prototype, thus restricting our ability to conduct a full test. However, we were able to perform a series of partial tests, which provided valuable insights into specific functionalities and potential applications of the device. This partial dataset contributes to our understanding and development of the technology

5.1 Description of the "product" with references to theory



Figure 22: Product case

The representation of our product, as illustrated in Figure 22 depicts the completed casing along with the integral components. The cable visible in Figure 22 leads to the probe, which consists of three LED diodes arranged in a triangular configuration with a single photodiode centrally located. In alignment with our objective to measure oxy- and deoxyhemoglobin levels, the distance between each LED and the photodiode is set at 8 mm. As described in section 3.6.2, the penetration depth of our chosen wavelength is directly influenced by this spacing. Specifically, the effective depth of the diode is calculated as half



Figure 23: Open case

the distance between the photodiode and the adjacent LED, resulting in a target penetration depth of approximately 4mm, which aligns with our design specifications. The device incorporates dual operational modes—digital and analog. Encased within a single unit, this dual-modality design not only optimizes space efficiency but also provides enhanced operational flexibility, allowing for seamless transitions between the analog and digital configurations, as demonstrated in Figure 23. The Arduino Mega 2560 serves as the core component of our device, strategically integrated within the casing to enhance usability and flexibility. For data transfer, a USB 2.0 cable of type A/B is discreetly routed from beneath the case to the computer, facilitating seamless digital communication. In terms of analog output, cables are strategically mounted to the side of the case, enabling easy connectivity to an oscilloscope and allowing for operation without necessitating an open case.

Additionally, the probe of our device is encased in a specially designed square box to avoid exposure and ensure durability. This casing is crafted from medical-grade silicone, chosen for its suitability in ensuring the probe’s effective application for the intended use-case. The process of casting this casing for the probe is depicted in Figure 24, illustrating the meticulous attention to detail in the device’s construction.

5.2 Experimental data

Had the device operated as anticipated, the expected data acquisition would have included baseline tissue oxygenation levels near the sural nerve at rest, serving as a control for subsequent measurements. Additionally, alterations in blood oxygenation in response to mechanical stimuli would have been observed,

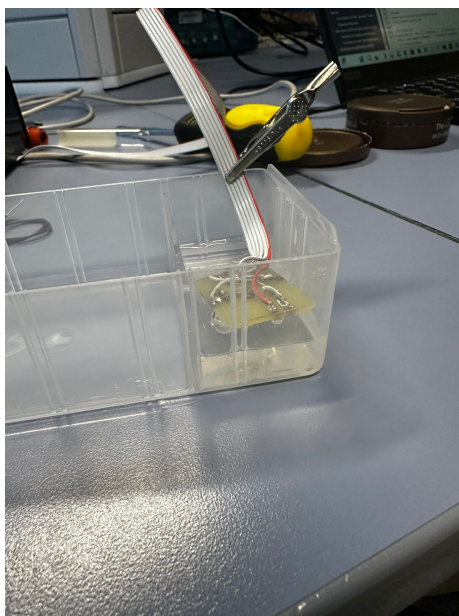


Figure 24: Casting of silicone case for probe

as evidenced by changes in near-infrared (NIR) light absorption and reflection patterns. Temporal data would have depicted the rapidity of vascular responses following nerve activation, providing insights into the dynamics of blood flow and nerve activation, thus affirming the device's capacity to track physiological changes.

Although the prototype did not perform as projected, the developmental and testing phases yielded significant insights into the complexities of measuring deep muscle activity using NIRS. The identification of potential sources of error during these phases will inform future enhancements to the device's design and operational methodologies.

During the development phase, a series of tests were conducted on our circuit. These tests revealed that the device was partially operational. Specifically, during the assembly and testing of our analog circuit, evaluations were carried out to verify the functionality of our filters, as depicted in figure 25 and 26. The tests were executed utilizing a 1Hz cut-off frequency for our second-order Sallen-Key low-pass filter. According to our assessments, this setting was validated as correct. At the same time, tests were performed to evaluate the efficacy of our high-pass filter, with the cut-off frequency for the second-order Sallen-Key high-pass filter established at 0.3Hz. Based on our parameters, the performance of these filters was deemed satisfactory.

In the development phase of the LED drivers, a series of evaluations were conducted to verify their sufficiency for the specified parameters. These tests were carried out prior to the development of a PCB circuit that integrates all

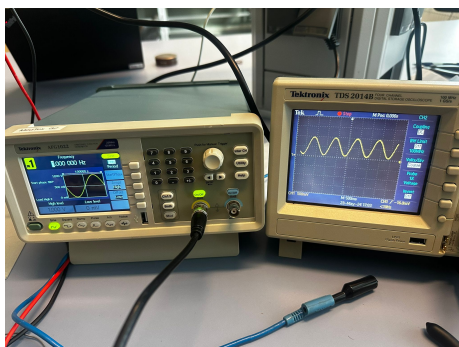


Figure 25: Using 5Hz simulated signal

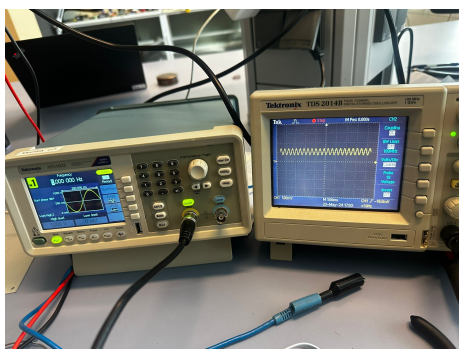


Figure 26: Same circuit filtering simulated signal in figure 25

the drivers, ensuring their functionality. The LED drivers were assembled on a breadboard and interfaced with our Arduino to simulate the intended use-case scenario, as illustrated in Figure 27

5.3 Discussion

At the start of this project, our focus was on acquiring comprehensive knowledge about the sural nerve and the methods for measuring muscle activity. After several months of research, we obtained the necessary understanding to begin designing our device. Upon completing the design, we proceeded with the assembly process, leaving us minimal time for troubleshooting. This was a significant oversight; earlier ordering of parts would have allowed ample time for testing and troubleshooting. Consequently, we faced major problems as we were unable to get the device fully functional. A structured plan and clear distribution of responsibilities would have been beneficial. With no time left for troubleshooting, we encountered two significant errors during testing. Firstly, LIA we attempted to program did not function correctly. After considerable

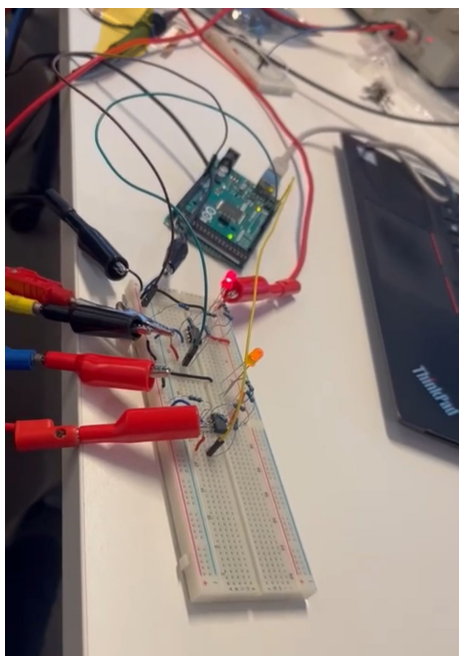


Figure 27: Test conducted for functionality of LED drivers

effort, we identified that the Arduino's sampling frequency was insufficient for the LIA's requirements. Secondly, while the LED drivers worked individually on the breadboard, they failed when assembled on the PCB and integrated with the LIA code. One LED received its signal correctly, but the other remained lit continuously, not blinking as intended. This issue did not occur when the LIA code was excluded, indicating a potential bug or a PCB-related problem. Despite testing all PCB connections with a voltmeter, the issue might have arisen during or after assembly, leaving us with two potential errors: a code bug and/or a PCB fault.

6 Conclusions and Further Work

The multifaceted and intricate nature of the task at hand, despite its focus on a minuscule segment of the human body, demands a comprehensive understanding of the physiological processes influenced by the sural nerve's activation. This report lays the groundwork necessary to intertwine the anatomy and physiology of the sural nerve with biomedical engineering. Subsequent enhancements can assess the alignment of theory with practical application. Expanding the project to measure the Cytochrome AA3 enzyme, related to metabolic activity at the tissue level, could offer further refinement of the anatomical and physiological dimensions of the project.

Signal processing improvements in the final device could be a game-changer, potentially unlocking numerous opportunities for the deployment of the measuring device. The integration of a digital lock-in amplifier could potentially solve a significant portion of the task. To concurrently measure Cytochrome AA3 and the oxygenation and deoxygenation levels of the tissue, the addition of a third LED with a shorter wavelength is proposed, possibly around 680nm. This necessitates enhancements to the microcontroller code managing the LEDs.

The optimization of the analog circuit and electronic components is required to augment the device's functionality. The potential application of filter techniques could analogically resolve the task, measuring the pulsatile signal (AC) without initial DC offset filtering. Afterward, filtering the DC component and measuring it, then subtracting the second signal from the first to compute the DC component. This method would necessitate higher amplification through the circuit than currently present. To incorporate a third LED for measuring the tissue's metabolism-related enzyme, an additional summing amplifier link should be introduced.

Product design is an area ripe for development within this project, with vast potential for enhancement. The device could potentially be more compact and user-friendly. 3D printing provides myriad possibilities for fabricating a suitable case for the LEDs and photodiode, which will be affixed to the skin. The creation of a well-designed ankle band is also a necessity.

Ultimately, this device could be amalgamated with other projects studying the sural nerve, contributing to human health. The objective is to observe the impact of the sural nerve's activation on the brain and body as a whole, and consequently, to discover the most effective methods for individuals to utilize their lower extremities.

References

- Abay, T. Y., & Kyriacou, P. A. (2022). Photoplethysmography in oxygenation and blood volume measurements. In J. Allen & P. Kyriacou (Eds.), *Photoplethysmography* (pp. 147-188). Academic Press. <https://doi.org/10.1016/B978-0-12-823374-0.00003-7>
- Abay, T. Y., & Kyriacou, P. A. (2015). Reflectance Photoplethysmography as Noninvasive Monitoring of Tissue Blood Perfusion. *IEEE Transactions on Biomedical Engineering*, 62(9), 2187-2195. <https://doi.org/10.1109/TBME.2015.2417863>
- Adams, W. (1942). The blood supply of nerves I: Historical review. *Journal of Anatomy*, 76(Pt 4), 323-341. Accessed February 7, 2024. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1252671/pdf/janat00513-0003.pdf>
- Althobaiti, M. (2023). Estimation of the Differential Pathlength Factor for Human Skin Using Monte Carlo Simulations. *Diagnostics (Basel, Switzerland)*, 13(2), 309. <https://doi.org/10.3390/diagnostics13020309>
- Alwazzan, M. (2020). Low cost blood vein detection system based on near-infrared LEDs and image-processing techniques. *Polish Journal of Medical Physics and Engineering*, 26, 61-67. <https://doi.org/10.2478/pjmpe-2020-0007>
- Arduino. (n.d.). Arduino Mega 2560 Rev3. <https://store.arduino.cc/products/arduino-mega-2560-rev3>
- Artinis Medical Systems. (2022, October 6). *NIRS explained: Absorbance & transmittance (Part 1/2)* [Video]. YouTube. <https://www.youtube.com/watch?v=vemBboFoyns>
- Carovac, A., Smajlovic, F., & Junuzovic, D. (2011). Application of ultrasound in medicine. *Acta Informatica Medica: AIM: Journal of the Society for Medical Informatics of Bosnia & Herzegovina: Časopis Društva za Medicinsku Informatiku BiH*, 19(3), 168-171. <https://doi.org/10.5455/aim.2011.19.168-171>
- Chawla, J., & Sahni, A. S. (2016). Peripheral nervous system anatomy. In T. R. Gest (Ed.), Medscape. <https://emedicine.medscape.com/article/1948687-overview?form=fpf>
- Cheriyedath, S. (2019, February 27). Photoplethysmography (PPG). News-Medical. [https://www.news-medical.net/health/Photoplethysmography-\(PPG\).aspx](https://www.news-medical.net/health/Photoplethysmography-(PPG).aspx)
- Cleveland Clinic. (2022). What is your sural nerve? <https://my.clevelandclinic.org/health/body/22323-sural-nerve>
- Dalen, H., & Brabrand, K. (2016). Apparatur - fysiske prinsipper. In S. Aakhus, K. Brabrand, H. Dalen, & T. Husebye (Eds.), *Medisisk ultralyddiagnostikk: Klinisk basisbok* (pp. 9-21). Gyldendal Akademisk.

- Ferrari, M., & Quaresima, V. (2012). A brief review on the history of human functional near-infrared spectroscopy (fNIRS) development and fields of application. *NeuroImage*, 63(2), 921-935. <https://doi.org/10.1016/j.neuroimage.2012.03.049>
- Fredriksson, I., Fors, C., Johansson, J., & Swiontkowski, M. F. (2012). Laser Doppler flowmetry – A theoretical framework. *ResearchGate*. <https://www.researchgate.net/publication/238678169>
- Fredriksson, I., Larsson, M., & Strömberg, T. (2009). Measurement depth and volume in laser Doppler flowmetry. *Microvascular Research*, 78(1), 4-13. <https://doi.org/10.1016/j.mvr.2009.02.008>
- Gao, Y., Weng, C., & Wang, X. (2013). Changes in nerve microcirculation following peripheral nerve compression. *Neural Regeneration Research*, 8(11), 1041-1047. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4145887/pdf/NRR-8-1041.pdf>
- Grose, K. (2022). Sural nerve: What to know. *WebMD*. <https://www.webmd.com/brain/sural-nerve-what-to-know>
- Handler, A., & Ginty, D. D. (2021). The mechanosensory neurons of touch and their mechanisms of activation. *Nature Reviews Neuroscience*, 22(9), 521-537. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8485761/pdf/nihms-1740144.pdf>
- Koga, S., Barstow, T., Okushima, D., Rossiter, H., Kondo, N., Ohmae, E., & Poole, D. (2015). Validation of a high-power, time-resolved, near-infrared spectroscopy system for measurement of superficial and deep muscle deoxygenation during exercise. *Journal of Applied Physiology*, 118(11), 1435-1442. <https://doi.org/10.1152/jappphysiol.01003.2014>
- Kremkau, F. W. (2020). *Sonography principles and instruments e-book* (10th ed.). Saunders. ISBN 9780323597098
- Kyriacou, P. A., Chatterjee, S. (2022). The origin of photoplethysmography. In J. Allen P. Kyriacou (Eds.), *Photoplethysmography* (pp. 17-43). Academic Press. <https://doi.org/10.1016/B978-0-12-823374-0.00004-9>
- Lee, Y., & Hwang, K. (2002). Skin thickness of Korean adults. *Surgical and Radiologic Anatomy*, 24(3-4), 183-189. <https://doi.org/10.1007/s00276-002-0034-5>
- Metzger, Z., Basrani, B., & Goodis, H. E. (2011). Instruments, materials, and devices. In K. M. Hargreaves & S. Cohen (Eds.), *Cohen's pathways of the pulp* (10th ed., pp. 223-282). Mosby. <https://doi.org/10.1016/B978-0-323-06489-7.00008-4>
- Miniato, M. A., & Nedeff, N. (2023, January). Anatomy, bony pelvis and lower limb: Sural nerve. *StatPearls - NCBI Bookshelf*. National Center for Biotechnology Information. <https://www.ncbi.nlm.nih.gov/books/NBK546638/>

- Nimana, K. V. H., Senevirathne, A. M. D. S. R. U., Pirannavan, R., et al. (2023). Anatomical landmarks for ankle block. *Journal of Orthopaedic Surgery and Research*, 18, 665. <https://doi.org/10.1186/s13018-023-04039-2>
- Oglat, A. A., Matjafri, M. Z., Suardi, N., Oqlat, M. A., Abdelrahman, M. A., & Oqlat, A. A. (2018). A review of medical Doppler ultrasonography of blood flow in general and especially in common carotid artery. *Journal of Medical Ultrasound*, 26(1), 3-13. https://doi.org/10.4103/JMU.JMU_11_17
- Okawa Electric Design. (n.d.). Digital filter design tool. Retrieved from <http://sim.okawa-denshi.jp/en/Fkeisan.htm>
- Oliveira, R., Semedo, S., Figueiras, E., Ferreira, L. F. R., & Humeau, A. (2011). Laser Doppler flowmeters for microcirculation measurements. In *Proceedings of the 1st Portuguese Biomedical Engineering Meeting* (pp. 1-4). Lisbon, Portugal. <https://doi.org/10.1109/ENBENG.2011.6026037>
- Owen-Reece, H., Smith, M., Elwell, C. E., & Goldstone, J. C. (1999). Near infrared spectroscopy. *BJA: British Journal of Anaesthesia*, 82(3), 418-426. <https://doi.org/10.1093/bja/82.3.418>
- Petricka, J. (n.d.). *An introduction to the lock-in amplifier and a Mie scattering experiment: 3 week sequence, for Gustavus Adolphus College PHY-305 (Junior advanced lab course)*. Gustavus Adolphus College. <https://www.compadre.org/Repository/document/ServeFile.cfm?ID=13827&DocID=4245>
- Proakis, J. G., Manolakis, D. G. (1996). Digital signal processing: Principles, algorithms, and applications (3rd ed.). Prentice-Hall. ISBN: 0-13-394338-9.
- Saha, M., Dremin, V., Rafailov, I., Dunaev, A., Sokolovski, S., & Rafailov, E. (2020). Wearable laser Doppler flowmetry sensor: A feasibility study with smoker and non-smoker volunteers. *Biosensors*, 10(12), 201. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7762214/pdf/biosensors-10-00201.pdf>
- Salzer, I., & Boehm, S. (2019). Regulation of nociceptor signaling by serotonin. In P. M. Pilowsky (Ed.), *Serotonin* (pp. 271-303). Academic Press. <https://doi.org/10.1016/B978-0-12-800050-2.00014-0>
- Scholkman, F., Kleiser, S., Metz, A. J., Zimmermann, R., Mata Pavia, J., Wolf, U., & Wolf, M. (2014). A review on continuous wave functional near-infrared spectroscopy and imaging instrumentation and methodology. *NeuroImage*, 85(1), 6-27. <https://doi.org/10.1016/j.neuroimage.2013.05.004>
- Shah, S., Price, D., Bukhman, G., Shah, S., & Wroe, E. (2011). Fundamentals. In *Manual of Ultrasound for Resource-Limited Settings* (pp. 21-36). Partners in Health. Retrieved from https://www.pih.org/sites/default/files/2017-07/6e013074d8f4c4c7d8_mlbfbxb8q.pdf
- Shetty, P. K., Galeffi, F., & Turner, D. A. (2012). Cellular links between neuronal activity and energy homeostasis. *Frontiers in Pharmacology*, 3. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3308331/pdf/fphar-03-00043.pdf>

- Smith, Y. (2019). What is subcutaneous tissue? News-Medical. <https://www.news-medical.net/health/What-is-Subcutaneous-Tissue.aspx>
- Sorbellini, E., Rucco, M., Rinaldi, F. (2018). Photodynamic and photobiological effects of light-emitting diode (LED) therapy in dermatological disease: an update. *Lasers in Medical Science*, 33, 1-9. <https://doi.org/10.1007/s10103-018-2584-8>
- Sviridova, N., Zhao, T., Aihara, K., Nakamura, K., & Nakano, A. (2018). Photoplethysmogram at green light: Where does chaos arise from? *Chaos, Solitons & Fractals*, 116, 157-165. <https://doi.org/10.1016/j.chaos.2018.09.016>
- Szczot, M., Nickolls, A. R., Lam, R. M., & Chesler, A. T. (2021). The form and function of PIEZO2. *Annual Review of Biochemistry*, 90, 507–534. <https://doi.org/10.1146/annurev-biochem-081720-023244>
- Yanagisawa, N., Koshiyama, M., Watanabe, Y., Sato, S., & Sakamoto, S. I. (2019). A quantitative method to measure skin thickness in leg edema in pregnant women using B-scan portable ultrasonography: A comparison between obese and non-obese women. *Medical Science Monitor*, 25, 1-9. <https://doi.org/10.12659/MSM.911799>